STAT332: Assignment 1

Ashley Chui (21003047)

January 2026

1. Let's consider a very small finite population U consisting of the following N = 6 units:

$$y_1 = 98 \quad y_2 = 102 \quad y_3 = 154 \quad y_4 = 133 \quad y_5 = 190 \quad y_6 = 175$$

From that population, we draw a random sample of size $n = 3$ using either sampling design I or sampling design II.

(a) For sampling design I, compute:

(i) $E(\bar{y_\pi})$

Since each unit appears in 4/8 samples, then $\pi_k = 4/8 = 1/2$. Then:

$$\bar{y_\pi} = \frac{1}{N}\sum_{k\in S}\frac{y_k}{\pi_k} = \frac{1}{6}\sum_{k\in S}\frac{y_k}{1/2} = \frac{1}{3}\sum_{k\in S} y_k$$

so that

$$\begin{aligned}
E(\bar{y_\pi}) &= \frac{1}{8}\sum_{i=1}^{8}\bar{y_\pi}(s_i) \\
&= \frac{1}{24}[(y_1+y_3+y_5)+(y_1+y_3+y_6)+(y_1+y_4+y_5)+(y_1+y_4+y_6)+(y_2+y_3+y_5) \\
&+ (y_2+y_3+y_5)+(y_2+y_3+y_6)+(y_2+y_4+y_5)+(y_2+y_4+y_6)] \\
&= \frac{1}{24}(4y_1+4y_2+4y_3+4y_4+4y_5+4y_6) \\
&= \frac{1}{6}(98+102+154+133+190+175) \\
&= 142.0
\end{aligned}$$

(ii) $V(\bar{y_\pi})$

Using the formula $V(\bar{y_\pi}) = E(\bar{y_\pi}^2) - E(\bar{y_\pi})^2$. We first calculate $E(\bar{y_\pi}^2)$:

$$\begin{aligned}
E(\bar{y_\pi}^2) &= \frac{1}{8}\sum_{i=1}^{8}\bar{y_\pi}^2(s_i) \\
&= \frac{1}{72}[(y_1+y_3+y_5)^2+(y_1+y_3+y_6)^2+(y_1+y_4+y_5)^2+(y_1+y_4+y_6)^2+(y_2+y_3+y_5)^2 \\
&+ (y_2+y_3+y_5)^2+(y_2+y_3+y_6)^2+(y_2+y_4+y_5)^2+(y_2+y_4+y_6)^2] \\
&= \frac{1}{72}(442^2+427^2+421^2+406^2+446^2+431^2+425^2+410^2) \\
&= \frac{1453172}{72}
\end{aligned}$$

so $V(\bar{y_\pi}) = \frac{1453172}{72} - 142^2 \approx 18.944$

(iii) $Bias(\bar{y_\pi})$

$$Bias(\bar{y_\pi}) = E(\bar{y_\pi}) - \bar{y} = 142 - 142 = 0$$

so $\bar{y_\pi}$ is an unbiased estimator as expected.

(iv) $MSE(\bar{y_\pi})$

$$MSE(\bar{y_\pi}) = V(\bar{y_\pi}) + Bias(\bar{y_\pi})^2 = 18.944 + 0^2 = 18.944$$

where MSE is the mean square error.
Hint: Though you can use

$$V(\hat{t_\pi}) = \sum_{k \in U} \sum_{l \in U} (\pi_{k,l} - \pi_k \pi_l) \frac{y_k}{\pi_k} \frac{y_l}{\pi_l}$$

(b) Same as (a), but with sampling design II.

(i) Start by computing π_k

$$\pi_1 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \quad \pi_2 = \frac{1}{2} \quad \pi_3 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \quad + \pi_4 = \frac{1}{4} \quad \pi_5 = \frac{1}{4} \quad \pi_6 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

Then we compute $\bar{y_\pi}$ for each possible sample using the same formula as in part (a):

$$II1 : \bar{y_\pi} = \frac{1}{6}(\frac{98}{1/2} + \frac{133}{1/4} + \frac{175}{3/4}) = \frac{1442}{9}$$
$$II2 : \bar{y_\pi} = \frac{1}{6}(\frac{102}{1/2} + \frac{154}{3/4} + \frac{175}{3/4}) = \frac{964}{9}$$
$$II3 : \bar{y_\pi} = \frac{1}{6}(\frac{98}{1/2} + \frac{154}{3/4} + \frac{190}{1/4}) = \frac{1742}{9}$$

Then $E(\bar{y_\pi}) = (\frac{1}{4})(\frac{1442}{9}) + (\frac{1}{2})(\frac{964}{9}) + (\frac{1}{4})(\frac{1742}{9}) = 142.0$

(ii) Using the formula $V(\bar{y_\pi}) = E(\bar{y_\pi}^2) - E(\bar{y_\pi})^2$. We first calculate $E(\bar{y_\pi}^2)$:

$$E(\bar{y_\pi}^2) = (\frac{1}{4})(\frac{1442}{9})^2 + (\frac{1}{2})(\frac{964}{9})^2 + (\frac{1}{4})(\frac{1742}{9})^2 = \frac{1743130}{81}$$

then $V(\bar{y_\pi}) = \frac{1743130}{81} - 142^2 \approx 1356.123$

(iii) $Bias(\bar{y_\pi}) = E(\bar{y_\pi}) - \bar{y} = 142 - 142 = 0$

(iv) $MSE(\bar{y_\pi}) = V(\bar{y_\pi}) + Bias(\bar{y_\pi})^2 = 1356.123$

(c) Which sampling design/plan do you think is better? Why? Although both sampling designs yield the same expected value and are both unbiased, sampling design I has a much smaller variance. Therefore, estimates from design I are far more precise and reliable than those from design II.

Sample #	S	$p(S)$
I1	$\{1,3,5\}$	1/8
I2	$\{1,3,6\}$	1/8
I3	$\{1,4,5\}$	1/8
I4	$\{1,4,6\}$	1/8
I5	$\{2,3,5\}$	1/8
I6	$\{2,3,6\}$	1/8
I7	$\{2,4,5\}$	1/8
I8	$\{2,4,6\}$	1/8

Table 1: Sampling design I (8 possible samples may be chosen).

Sample #	S	$p(S)$
II1	$\{1,4,6\}$	1/4
II2	$\{2,3,6\}$	1/2
II3	$\{1,3,5\}$	1/4

Table 2: Sampling design II (3 possible samples may be chosen).

2. (a) As mentioned in class $\hat{p_S}$ is a special case of $\bar{y_S}$ (where the y_k's can only take two values 0 and 1). Show that when estimating a proportion (under SRS), the variance estimator

$$\hat{V}_{SRS}(\hat{p_S}) = \hat{V}_{SRS}(\hat{\bar{y_S}}) = (\frac{1}{n} - \frac{1}{N})S_S^2$$

simplifies to

$$\hat{V}_{SRS}(\hat{p_S}) = (\frac{1}{n} - \frac{1}{N})\frac{n}{n-1}\hat{p_S}(1-\hat{p_S})$$

In other words, show that formula (2.10) of the slides simplifies to (2.20), when estimating a proportion.

Start by showing that the sample variance $S_S^2 = \frac{1}{n-1}(n\hat{p_S})(1-\hat{p_S})$.

$$\begin{aligned}
S_S^2 &= \frac{1}{n-1}\sum_{k\in S}(y_k - \bar{y_S})^2 \\
&= \frac{1}{n-1}(\sum y_k^2 - 2\bar{y_S}\sum y_k + \bar{y_S}^2\sum 1 \qquad \text{where } \sum y_k^2 = \sum y_k \text{ since } p_k = y_k \in 0,1 \\
&= \frac{1}{n-1}(n\hat{p_S} - 2n\hat{p_S}^2 + n\hat{p_S}^2) \\
&= \frac{1}{n-1}(n\hat{p_S})(1-\hat{p_S})
\end{aligned}$$

Then

$$\hat{V}_{SRS}(\hat{p_S}) = (\frac{1}{n} - \frac{1}{N})\frac{n}{n-1}\hat{p_S}(1-\hat{p_S}) \quad \text{as needed.}$$

(b) An alternate estimator of $V_{SRS}(\hat{p_S})$ is obtained by simply replacing p_U by $\hat{p_S}$ in (2.18); thus obtaining

$$\hat{V^*_{SRS}} = (\frac{1}{n} - \frac{1}{N}\frac{N}{N-1}\hat{p_S}(1-\hat{p_S})$$

Recalling that $\hat{V_{SRS}}$ is an unbiased estimator of V_{SRS}, compute the expected value of $\hat{V^*_{SRS}}$; thus showing that $\hat{V^*_{SRS}}$ is biased.

$$E(\hat{V^*_{SRS}}(\hat{p_S})) = (\frac{1}{n} - \frac{1}{N})\frac{N}{N-1}E(\hat{p_S}(1-\hat{p_S}) \tag{1}$$

where

$$\begin{aligned}
E(\hat{p_S}(1-\hat{p_S}) &= E(\hat{p_S}) - E(\hat{p_S}^2) \quad \text{where } E(\hat{p_S}) = p_U \text{ since } \hat{p_S} \text{ is an unbiased estimator for } p_U \\
&= p_U - (V(\hat{p_S}) + E(\hat{p_S}^2)) \qquad \text{since } E(X^2) = V(X) + E(X)^2 \\
&= p_U - p_U^2 - V(\hat{p_S}) \\
&= p_U(1-p_U) - (\frac{1}{n} - \frac{1}{N})\frac{N}{N-1}p_U(1-p_U) \qquad \text{by eq. (2.18)} \\
&= p_U(1-p_U)(1-A) \qquad \text{where } A = (\frac{1}{n} - \frac{1}{N})\frac{N}{N-1}
\end{aligned}$$

Therefore, eq (1) becomes $E(\hat{V^*_{SRS}}(\hat{p_S})) = A(1-A)p_U(1-p_U)$. Whereas, the variance is $V_{SRS}(\hat{p_S}) = Ap_U(1-p_U)$. Hence,

$$\begin{aligned}
Bias(\hat{V^*_{SRS}}(\hat{p_S})) &= E(\hat{V^*_{SRS}}(\hat{p_S})) - V_{SRS}(\hat{p_S}) \\
&= A(1-A)p_U(1-p_U) - Ap_U(1-p_U) \\
&= -A^2p_U(1-p_U)
\end{aligned}$$

since $E(\hat{V^*_{SRS}}(\hat{p_S})) \neq V_{SRS}(\hat{p_S})$ and hence $Bias(\hat{V^*_{SRS}}(\hat{p_S})) \neq 0$, then $\hat{V^*_{SRS}}(\hat{p_S})$ is a biased estimator of $V_{SRS}(\hat{p_S})$.

(c) What happens to the bias of $\hat{V_{SRS}^*}(\hat{p_S})$ computed in part b) when $n \to N$?
In part b),

$$Bias(\hat{V_{SRS}^*}(\hat{p_S})) = -((\frac{1}{n} - \frac{1}{N})\frac{N}{N-1})^2 p_U(1-p_U)$$

where $\frac{1}{n} - \frac{1}{N} \to 0$ as $n \to N$, and hence $Bias(\hat{V_{SRS}^*}(\hat{p_S})) \to 0^2 p_U(1-p_U) = 0$.
Therefore, as the sample becomes closer to the whole population, the bias goes to 0.

3. Recall that some sampling design $p(S)$ have a fixed sample size n (e.g. SRS), whereas others have a random sample size n_S (e.g. Bernoulli sampling). Prove that

$$V^*(\hat{t_\pi}) = \frac{1}{2}\sum_{k\in U}\sum_{l\in U}(\pi_{k,l} - \pi_k\pi_l)(\frac{y_k}{\pi_k} - \frac{y_l}{\pi_l})^2$$

is equivalent to

$$V(\hat{t_\pi}) = \sum_{k\in U}\sum_{l\in U}(\pi_{k,l} - \pi_k\pi_l)\frac{y_k}{\pi_k}\frac{y_l}{\pi_l}$$

for fixed size sampling designs.

Note that you are not allowed to assume any given sampling design (this includes SRS).

$$\begin{aligned}
V^*(\hat{t}_\pi) &= -\frac{1}{2}\sum_{k\in U}\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell)\left(\frac{y_k}{\pi_k} - \frac{y_\ell}{\pi_\ell}\right)^2 \\
&= -\frac{1}{2}\sum_{k\in U}\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell)\left(\left(\frac{y_k}{\pi_k}\right)^2 + \left(\frac{y_\ell}{\pi_\ell}\right)^2 - 2\frac{y_k}{\pi_k}\frac{y_\ell}{\pi_\ell}\right) \\
&= -\frac{1}{2}\left[\sum_{k\in U}\sum_{\ell\in U}(\pi_{k,\ell} - \pi_k\pi_\ell)\left(\frac{y_k}{\pi_k}\right)^2 + \sum_{k\in U}\sum_{\ell\in U}(\pi_{k,\ell} - \pi_k\pi_\ell)\left(\frac{y_\ell}{\pi_\ell}\right)^2 - 2\sum_{k\in U}\sum_{\ell\in U}(\pi_{k,\ell} - \pi_k\pi_\ell)\left(\frac{y_k}{\pi_k}\frac{y_\ell}{\pi_\ell}\right)\right]
\end{aligned}$$

The first two double sums are equal by symmetry of the indices of k and ℓ. Note that for fixed sample size n: $\sum_{\ell\in U} I_\ell = n$ and consider one of the sums:

$$\begin{aligned}
\sum_{k\in U}\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell)\left(\frac{y_k}{\pi_k}\right)^2 &= \sum_{k\in U}\left(\frac{y_k}{\pi_k}\right)^2\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell) \\
&= \sum_{k\in U}\left(\frac{y_k}{\pi_k}\right)^2\left[\sum_{\ell\in U}\pi_{k\ell} - \pi_k\sum_{\ell\in U}\pi_\ell\right] \\
&= \sum_{k\in U}\left(\frac{y_k}{\pi_k}\right)^2\left[\sum_{\ell\in U}E(I_kI_\ell) - \pi_k\sum_{\ell\in U}E(I_\ell)\right] \\
&= \sum_{k\in U}\left(\frac{y_k}{\pi_k}\right)^2[nE(I_k) - \pi_k n] \\
&= \sum_{k\in U}\left(\frac{y_k}{\pi_k}\right)^2(n\pi_k - \pi_k n) \\
&= 0
\end{aligned}$$

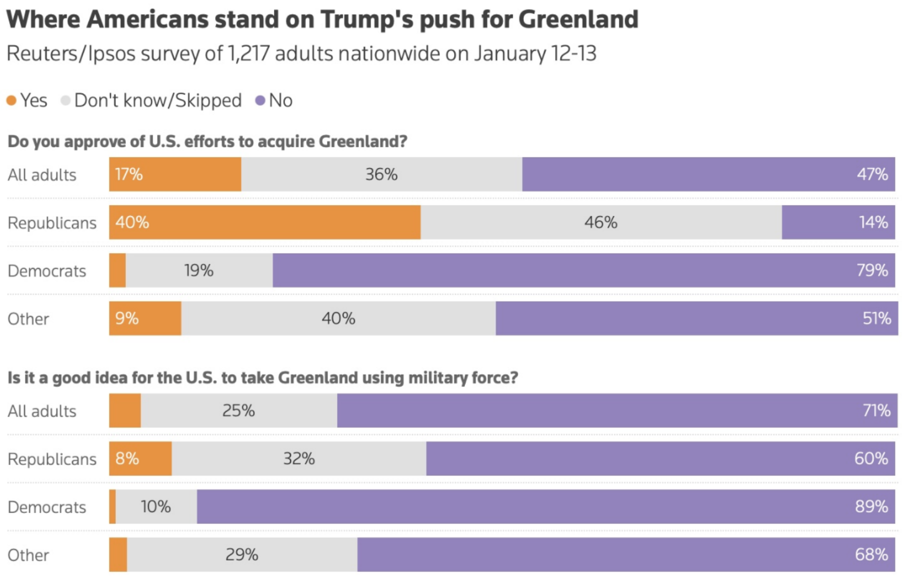
Hence both squared terms equal 0 and the equation above becomes

$$\begin{aligned}
V^*(\hat{t}_\pi) &= -\frac{1}{2}\left(-2\sum_{k\in U}\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell)\frac{y_k}{\pi_k}\frac{y_\ell}{\pi_\ell}\right) \\
&= \sum_{k\in U}\sum_{\ell\in U}(\pi_{k\ell} - \pi_k\pi_\ell)\frac{y_k}{\pi_k}\frac{y_\ell}{\pi_\ell}.
\end{aligned}$$

Therefore, for any fixed-size sampling design,

$$V^*(\hat{t}_\pi) = V(\hat{t}_\pi).$$

4. A very recent poll conducted by Reuters/Ipsos on Jan 12-13 found that only 17% of U.S. adults approve of President Donald Trump's efforts to acquire Greenland and only 4% think that it is a good idea for the U.S. to take Greenland using military force. This poll was conducted online using a national sample of 1,217 US adults. Though the actual methodology used by Ipsos is more complicated, for this question, let's simply assume that this sample was obtained via SRS.



(a) Provide a 95% confidence interval for the point estimate of 17%.

Assume the sample of size $n = 1217$ was obtained via SRS from a very large population ($N \gg n$), so $1/N \approx 0$.

The point estimate is $\hat{p}_S = 0.17$. From equation (2.21), the $100(1-\alpha)\%$ CI for the population proportion p_U is

$$p_U \in \hat{p}_S \pm t_{n-1,1-\alpha/2}\sqrt{\left(\frac{1}{n} - \frac{1}{N}\right)\frac{n}{n-1}\hat{p}_S(1-\hat{p}_S)}.$$

Since N is very large, $\frac{1}{N} \approx 0$, and because n is large, $t_{n-1,0.975} \approx 1.96$. Thus,

$$p_U \approx \hat{p}_S \pm 1.96\sqrt{\frac{\hat{p}_S(1-\hat{p}_S)}{n-1}}.$$

Substituting $\hat{p}_S = 0.17$ and $n = 1217$,

$$SE = \sqrt{\frac{0.17(0.83)}{1216}} \approx 0.01077,$$

Therefore, the 95% CI is (14.9%, 19.1%)

(b) Provide a 95% confidence interval for the point estimate of 4%.

Note that you will need to use CI for small proportions in part b, but not part a

Since 17% is considered a relatively small proportion of all U.S. adults, we will use the logit transformation method to obtain a CI.
Define

$$\text{logit}(\hat{p}_S) = \log\left(\frac{\hat{p}_S}{1-\hat{p}_S}\right) = \log\left(\frac{0.04}{0.96}\right) \approx -3.1781.$$

From the notes, the estimated SRS variance of $\hat{p}_S$ is

$$\widehat{V}_{\text{SRS}}(\hat{p}_S) = \left(\frac{1}{n} - \frac{1}{N}\right)\frac{n}{n-1}\hat{p}_S(1-\hat{p}_S) \approx \frac{\hat{p}_S(1-\hat{p}_S)}{n-1}.$$

Using the delta method,

$$\widehat{V}_{\text{SRS}}\big(\text{logit}(\hat{p}_S)\big) \approx \frac{\widehat{V}_{\text{SRS}}(\hat{p}_S)}{\big(\hat{p}_S(1-\hat{p}_S)\big)^2},$$

so

$$\widehat{SE}\big(\text{logit}(\hat{p}_S)\big) \approx 0.1463.$$

The logit-scale limits are

$$l, u = \text{logit}(\hat{p}_S) \pm t_{n-1,0.975}\,\widehat{SE}\big(\text{logit}(\hat{p}_S)\big) \approx -3.1781 \pm 1.96(0.1463),$$

giving

$$l \approx -3.4649, \qquad u \approx -2.8912.$$

By equation (2.22), the $100(1-\alpha)\%$ CI for p_U is

$$p_U \in \left[\frac{e^l}{1+e^l},\ \frac{e^u}{1+e^u}\right] = (0.0303,\ 0.0526).$$

Thus, the 95% logit CI is $(3.03\%,\ 5.26\%)$